Black Soldier Fly Larvae: - A cheaper option for chicken feed formulation

November 2025

Farmer Benefits and Evidence

Why Use Black Soldier Fly Larvae in Chicken Feeds? Black Soldier Fly Larvae (BSFL) provide a sustainable, high protein alternative to costly feeds such as soya bean and fish meal. By recycling farm and household waste into valuable protein, BSFL help farmers cut feed costs, maintain egg production, and improve household incomes.

Farmers using BSFL feeds cut costs and keep egg production steady

Benefits for Farmers

- **Lower feed costs:** BSFL meal reduces daily feed expenses.
- **Stable egg output:** No drop in production.

- **Higher profits:** More income gained.
- **Quick payback:** Savings offset setup costs rapidly.

Additional Benefits

- **Waste recycling:** BSFL turn farm waste into valuable feed.
- **Organic fertilizer:** Frass enriches soils with nitrogen and potassium.
- **Poultry feed:** Affordable protein for laying hens.
- **By-products:** Pupa and casings support biodiesel, cosmetics, and soil amendments.

BSFL-based feeds are practical, affordable, and environmentally friendly. By adopting BSF production, farmers secure reliable protein for chickens, reduce expenses, and strengthen household livelihoods.

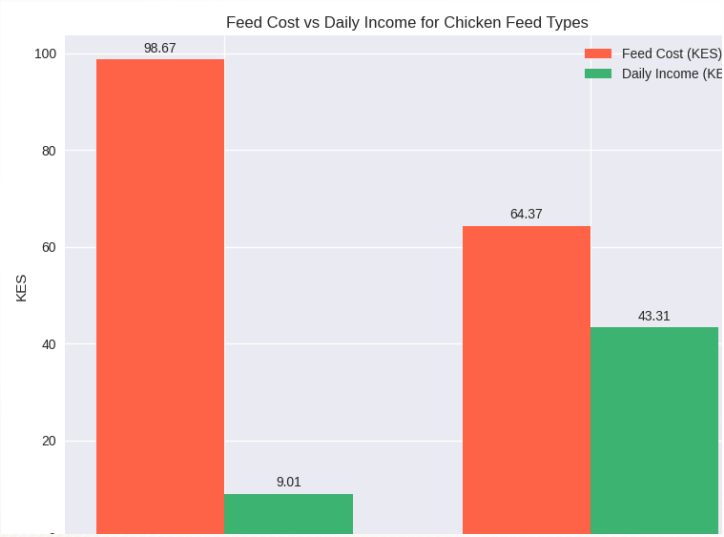
Community adoption of BSF strengthens poultry enterprises

Farmer group training session

> *Using BSFL feeds increases daily income by more than four times, mainly due to reduced feed costs.*

Evidence from Feeding Trials

Tests by KALRO on BSFL-based rations (B7–B10, plus B3) against a control ration using KC1 laying hens showed that BSFL-based rations were **19.5–34.8% cheaper** than the control ration, while meeting Kenya Bureau of Standards requirements for layer feeds. Farmers reported steady egg production and significant income gains.



BSFL rations are cheaper and maintain egg production.

> ***BSFL based Ration B3 cuts feed costs and boosts daily income more than four times.***

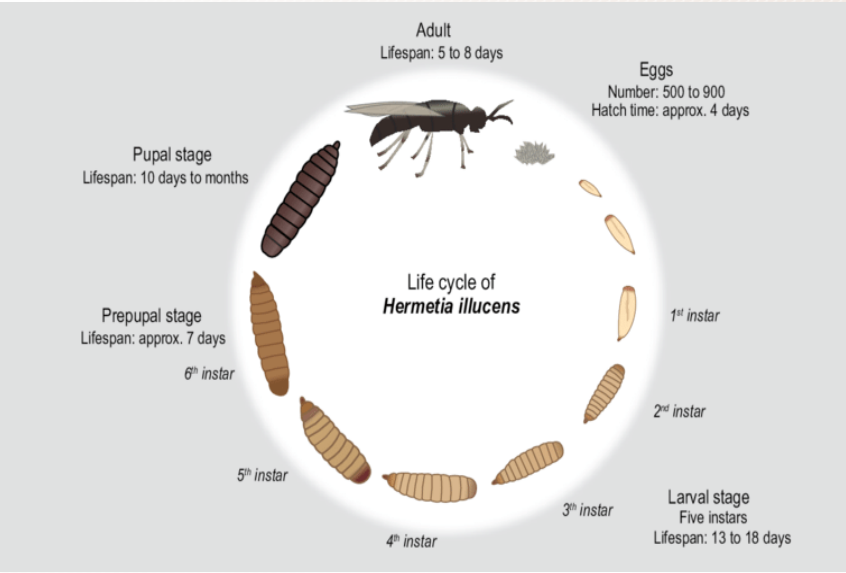
Commercial Layers Feed costs nearly KES 99/day for 10 hens, leaving only KES 9 income. BSFL ration B3 reduces feed costs to KES 64/day and raises income to over KES 43/day.

Technical Notes Life Cycle and Nutrients

The BSF life cycle includes egg, larval, prepupal, pupal, and adult stages. Under ideal conditions of temperature and humidity, the cycle completes in about 45 days, ensuring a steady supply of larvae for feed. At the prepupal stage, they contain:

- 42% crude protein
- 35% fat
- 5% calcium
- 1.5% phosphorus

Production costs average **KES 130/kg**, making BSFL meal cheaper than imported protein sources.



The life cycle of Black Soldier Fly

The BSF life cycle takes about 45 days under ideal screen house conditions

"Larvae recycle waste into protein for poultry feed."

Highquality nutrients at only KES 130/kg

"Turn waste into feed, feed into profit.

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